# Setting Type Whole

*A troff for InferNode*

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InferNode Project

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# Lines

# A line at a time

Troff fills a line until the next word will not fit, then adjusts it [1]

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- A loose line is found only when it is set
- The lines before it could have helped, and are gone
- Knuth and Plass choose every break of a paragraph at once [2]

# A paragraph broken whole

- Each break a node; each line between two an edge
- The paragraph the shortest path through them
- A hyphen costs as a loose line does

$$d = (l + b)^2 + p^2$$

A line's demerits: $l$ a constant, $b$ the line's badness, $p$ the penalty of the break it ends at

# Mathematics

# Mathematics as it is spoken

Kernighan and Cherry's language [3] set by TeX's rules [4, ch. 17]

```
x = {- b +- sqrt {b sup 2 - 4 a c}}
    over {2 a}
```

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

And within a line: $e^{i\pi} + 1 = 0$.

# Measured against LaTeX

| | troff | XeLaTeX | LuaLaTeX |
|---|---|---|---|
| A chapter, 3 pages | 0.7 s | 5.9 s | 9.5 s |
| A dissertation, 235 pages | 12.4 s | 42.6 s | 62.8 s |
| Mathematics, 245 pages | 13.9 s | 4.6 s | 11.0 s |

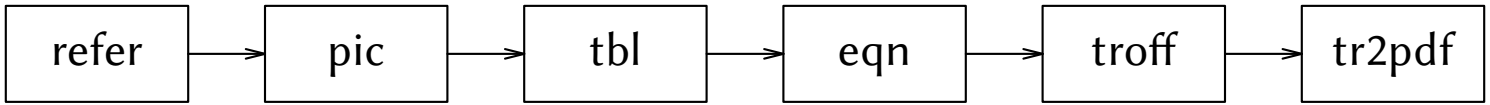
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Where mathematics fills every page, TeX is faster: troff measures a formula's parts by setting them again.

# Slides

# The pipeline




`book -m slides` runs those a talk needs, as for a book; pic drew this [5]

# A slide, written

```
.slide "A line at a time"
Troff fills a line until ...
.pause
.item
A loose line is found only when it is set
```

- Each `.pause` shows the slide again, with more
- What is shown already does not move

# References


[1] J. F. Ossanna, “NROFF/TROFF User’s Manual,” Bell Laboratories, Murray Hill, N.J., Computing Science Technical Report 54, 1976.

[2] D. E. Knuth and M. F. Plass, “Breaking Paragraphs into Lines,” *Software: Practice and Experience*, vol. 11, no. 11, pp. 1119–1184, 1981.

[3] B. W. Kernighan and L. L. Cherry, “A System for Typesetting Mathematics,” *Communications of the ACM*, vol. 18, no. 3, pp. 151–157, 1975.

[4] D. E. Knuth, *The TeXbook*. Reading, Mass.: Addison-Wesley, 1984.

[5] B. W. Kernighan, “PIC: A Language for Typesetting Graphics,” *Software: Practice and Experience*, vol. 12, no. 1, pp. 1–21, 1982.